Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Enterprise Java

Practical – IV

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Class : TYIT	Date/Time of Submission : 18/07/2026

AIM(A): Develop a simple JSP application to display values obtained from the use of intrinsic objects of various types.

CODE:-

index.jsp:

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<html><head><title> JSP Page</title></head>
<body>
<h1>Use of Intrinsic Objects in JSP</h1>
<h1>Request Object </h1>
Query String <%=request.getQueryString() %><br>
Context Path <%=request.getContextPath() %><br>
Remote Host <%=request.getRemoteHost() %><br>
<h1>Response Object </h1>
Character Encoding Type <%=response.getCharacterEncoding() %><br>
Content Type
<%=response.getContentType() %><br>
Locale <%=response.getLocale() %><br>
<h1>Session Object </h1>
ID <%=session.getId() %><br>
Creation Time <%=new java.util.Date(session.getCreationTime()) %><br>
Last Access Time<%=new java.util.Date(session.getLastAccessedTime()) %><br>
</body>
</html>
```

OUTPUT:

Use of Intrinsic Objects in JSP

Request Object

Query String null
Context Path /WebApplication1
Remote Host 0:0:0:0:0:0:0:1

Response Object

Character Encoding Type UTF-8
Content Type text/html;charset=UTF-8
Locale en_IN

Session Object

ID 457a7ce865ed1a48aa52a6c3297e
Creation Time Sat Jul 18 13:58:49 IST 2026
Last Access TimeSat Jul 18 13:58:49 IST 2026

AIM(B): Develop a simple JSP application to pass values from one page to another with validations. (Name-txt, age-txt, hobbies-checkbox, email-txt, gender-radio button).

CODE:-

index.jsp:

```
<!DOCTYPE html>
<html>
 <head>
 <meta https-equiv="Content-Type" content="text/html;charset=UTF-8">
 <title>JSP Registration</title>
 </head>
 <body>
 <h1>Registration Form</h1>
 <form action="val.jsp" method="post">
 <table style="width:50%">
 <tr>
 <td>Full Name</td>
 <td><input type="text" name="fullname"/></td>
```

OUTPUT:

Registration Form

Full Name	omkar
Age	20
E-Mail	user1111@gmail.com
Gender	◉ Male ○ Female
Hobbies	☐ Acting ☐ Dancing ☐ Singing ☑ Drawing

Register

Your Extended Information is as follows:

Full Name:**omkar**
Age:**20**
EMail:**user1111@gmail.com**
Gender:**Male**
Hobbies: **Drawing**

```
</tr><tr>
<td>Age</td>
<td><input type="text" name="age"/></td>
</tr><tr>
<td>E-Mail</td>
<td><input type="text" name="email" size="20"/></td>
</tr><tr>
<td>Gender</td>
<td><input type="radio" name="gender" value="Male">Male
<input type="radio" name="gender" value="Female"/>Female
</td>
</tr><tr>
<td>Hobbies</td>
<td><input type="checkbox" name="hb" value="Acting"/>Acting
<input type="checkbox" name="hb" value="Dancing"/>Dancing
<input type="checkbox" name="hb" value="Singing"/>Singing
<input type="checkbox" name="hb" value="Drawing"/>Drawing
</td>
</tr>
</table>
<input type="submit" value="Register"/>
</form>
</body>
</html>
```

val.jsp:

```
<%@page language="java" contentType="text/html" pageEncoding="UTF-8" %>
<%!
 int ageInNumbers;
private static final String EMAIL_REGEX="^[\\w-\\+]+(\\.[\\w]+)*@[\\w-]+(\\.[\\w]+)*(\\.[a-z]{2,})$";
%>
<%
 String name=request.getParameter("fullname");
 String age=request.getParameter("age");
 String email=request.getParameter("email");
 String gender=request.getParameter("gender");
 String hb[]=request.getParameterValues("hb");
  if (name == null || name.isEmpty() || age == null || age.isEmpty() || email == null || email.isEmpty() || gender
==  null || gender.isEmpty()) {
 out.println("<font color=red>Please fill all the fields</font><br>");
 }
if(!email.matches(EMAIL_REGEX)){
 out.println("<font color=red>Correct Your Email Address</font><br>");
 }
try{
 ageInNumbers=Integer.parseInt(age.trim());
 }
catch(NumberFormatException e){
 out.println("<font color=red>Age must be numbers</font><br>");
 }
if (ageInNumbers < 18 || ageInNumbers > 60) {
 out.println("<font color=red>Age must be between 18 and 60</font><br>");
 }
%>
Your Extended Information is as follows:<br><br>
Full Name<b>:<%=name%></b><br>
Age<b>:<%=age%></b><br>
```

```
EMail<b>:<%=email%></b><br>
Gender<b>:<%=gender%></b><br>
Hobbies<b>:
 <%
 if(hb!=null && hb.length!=0){
 for(int i=0;i<hb.length;i++){
 out.println(hb[i]);
 }
 }
 %>
```

web.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee http://xmlns.jcp.org/xml/ns/javaee/web
app_3_1.xsd"
         version="3.1">
 <welcome-file-list>
 <welcome-file>index.jsp</welcome-file>
 </welcome-file-list>
</web-app>
```

AIM(C): Create a registration and login JSP application to register and authenticate the user based on username and password using JDBC

CODE:-

Register.jsp

```
<!DOCTYPE html>
<html>
   <head>
      <title>Registration</title>
   </head>
   <body>
      <h2>User Registration</h2>
      <form action="RegisterServlet" method="post">
         Username
         <input type="text" name="username"><br><br>
         Password
         <input type="password" name="password"><br><br>
         <input type="submit" value="Register">
      </form>
      <br>
      <a href="login.jsp">Already Registered? Login</a>
   </body>
</html>
```

Login.jsp

```
<!DOCTYPE html>
<html>
   <head>
      <title>Login</title>
   </head>
 <body>
<h2>User Login</h2>
<form action="LoginServlet" method="post">
         Username
         <input type="text" name="username"><br><br>
         Password
         <input type="password" name="password"><br><br>
```

OUTPUT:




```
mysql> CREATE DATABASE studentdb;
Query OK, 1 row affected (0.02 sec)

mysql>
mysql> USE studentdb;
Database changed
mysql>
mysql> CREATE TABLE users(
    -> id INT AUTO_INCREMENT PRIMARY KEY,
    -> username VARCHAR(50),
    -> password VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.98 sec)

mysql> SHOW DATABASES;
+--------------------+
```

```
        <input type="submit" value="Login">
      </form>
   </body>
</html>
```

LoginServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {
 @Override
   protected void doPost(HttpServletRequest request,
         HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      String username = request.getParameter("username");
      String password = request.getParameter("password");
      try {
       Connection con = DBConnection.getConnection();
if (con == null) {
   out.println("Database Connection Failed");
   return;
}
PreparedStatement ps = con.prepareStatement(
      "SELECT * FROM users WHERE username=? AND password=?");
         ps.setString(1, username);
         ps.setString(2, password);
         ResultSet rs = ps.executeQuery();
         if (rs.next()) {
            response.sendRedirect("success.jsp");
         } else {
            out.println("<h2>Invalid Username or Password</h2>");          }
      } catch (SQLException | IOException e) {
         out.println(e);
      }
   }
}
```

```
mysql> select * from users
    -> ;
+----+----------+----------+
| id | username | password |
+----+----------+----------+
|  1 | root     | root     |
+----+----------+----------+
1 row in set (0.00 sec)
```

RegisterServlet.java

```
import java.io.IOException;
import java.sql.Connection;
import java.sql.PreparedStatement;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
@WebServlet("/RegisterServlet")
public class RegisterServlet extends HttpServlet {

   protected void doPost(HttpServletRequest request,
         HttpServletResponse response)
         throws ServletException, IOException {
```

```
        String username = request.getParameter("username");
        String password = request.getParameter("password");

        try {

            Connection con = DBConnection.getConnection();

            PreparedStatement ps = con.prepareStatement(
                  "insert into users(username,password) values(?,?)");

            ps.setString(1, username);
            ps.setString(2, password);

            ps.executeUpdate();

            response.sendRedirect("login.jsp");

        } catch (Exception e) {

            e.printStackTrace();
        }

    }

}
```

success.jsp

```
<!DOCTYPE html>
<html>
    <head>
        <title>Success</title>
    </head>

    <body>

        <h1>Login Successful</h1>

        <h2>Welcome</h2>

    </body>
</html>
```